

Two-Phase Dynamics of Ammonia Emissions from Stored Pig Slurry: Interactions Between Nitrogen Transformations and Organic N Mineralization

Joonhee Lee ¹  and Heekwon Ahn ^{2,*} ¹ Department of Animal Science, Kyungpook National University, Sangju 37224, Republic of Korea; iamleejh@knu.ac.kr² Department of Animal Biosystems Science, Chungnam National University, Daejeon 34134, Republic of Korea

* Correspondence: hkahn@cnu.ac.kr; Tel.: +82-042-821-5785

Abstract

The temporal dynamics of nitrogen (N) fractions and ammonia (NH₃) volatilization were investigated over a 56-day storage period using a laboratory-scale pig slurry pit simulator. A detailed N mass balance, encompassing total N (TN), total ammonium N (TAN), organic N, and nitrate N (NO₃[−]-N) fractions, yielded a N mass recovery of 96.5%, despite uncertainties associated with discrete emission measurements, with a TN reduction of 28.3 g vessel^{−1} closely matched by cumulative NH₃-N emissions of 27.3 g. The NH₃ emission profile exhibited a distinct two-phase pattern. During Phase I (days 1–28), emissions remained stable at 16.7–19.5 g m^{−2} d^{−1}, accounting for approximately 58% of total cumulative NH₃-N loss (518.6 g m^{−2}), consistent with zero-order kinetics. Phase II (days 29–56) was characterized by first-order exponential decay ($k = 0.0293 \text{ d}^{-1}$, $R^2 = 0.982$), coinciding with progressive TAN depletion. Measured emission rates were strongly correlated with theoretical free ammonia N (FAN) concentrations derived from pH and temperature ($R^2 = 0.74$), confirming that theoretical FAN provides a useful upper bound for emission potential, although the actual gaseous flux is restricted by mass-transfer limitations at the slurry–air interface. These results demonstrate that continuous pH and temperature monitoring provides a practical basis for tracking emission dynamics and informing the timing of mitigation interventions, particularly during the high-flux initial storage phase.

Keywords: Nitrogen loss; ammonia emission; nitrogen mass balance; pig slurry; free ammonia; storage period



Academic Editors: Abele Kuipers and Adam Cieřlak

Received: 20 April 2026

Revised: 15 May 2026

Accepted: 21 May 2026

Published: 24 May 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

1. Introduction

Pig slurry is a heterogeneous mixture of feces, urine, and water, characterized by elevated concentrations of essential plant nutrients, particularly nitrogen (N) [1]. During storage in slurry pits, the slurry undergoes a series of physical, chemical, and biological transformations, including microbial decomposition of organic matter [2,3] and volatilization of N compounds [4–6].

Among the primary environmental concerns associated with slurry storage is the substantial loss of N, which proceeds through multiple pathways, most notably the volatilization of ammonia (NH₃) and nitrous oxide (N₂O) [6,7]. Elevated atmospheric concentrations of NH₃ pose direct health risks to livestock [8] and farm personnel [9], and serve as a significant precursor to secondary aerosol formation and atmospheric particulate matter (PM), an effect particularly pronounced in regions with high livestock density [10].

N₂O is a potent greenhouse gas with well-documented contributions to global warming and stratospheric ozone depletion [11]. Emissions originating from slurry pits beneath slatted floors have been estimated to account for 0.3–4.0% of total greenhouse gas emissions from pig houses [12]. The formation of N₂O proceeds through the microbial processes of nitrification and denitrification, for which the ammonium ion (NH₄⁺) serves as the primary substrate. Accordingly, gaseous NH₃ emission, aqueous NH₄⁺ concentration, and N₂O production are mechanistically coupled within the manure management system, underscoring the importance of understanding NH₃ dynamics as a prerequisite for broader N loss assessment.

NH₃ volatilization occurs when dissolved ammoniacal N is converted to free NH₃ gas and released into the atmosphere—a process that is particularly pronounced during the early stages of storage, when slurry pH tends to be high and ambient temperatures are elevated [7,13,14]. For example, at ambient temperatures of 9.2, 11.1, and 17.1°C, cumulative NH₃ emissions from slurry stored over two months reached 399, 1,116, and 1,593 g NH₃-N m^{−3} slurry, respectively [7]. Slurry acidification has demonstrated considerable efficacy in reducing these losses, with NH₃ emissions decreasing by 50%, 62%, and 77% at pH 6.0, 5.8, and 5.5, respectively, relative to non-acidified controls [15].

The equilibrium between free NH₃ (FA) and ionized ammonium (NH₄⁺) in solution is governed by pH and temperature, as described by the dissociation equilibrium of NH₃ in water [13,16,17]. Under acidic conditions, the elevated H⁺ concentration shifts the equilibrium toward NH₄⁺, whereas alkaline conditions favor the formation of FA, thereby increasing ammonia volatilization. Accurate characterization of this equilibrium is therefore central to quantifying N loss and evaluating the effectiveness of mitigation strategies.

Various management measures have been proposed to reduce N losses during pig slurry storage, including the installation of pit covers to limit surface exposure [18,19] and the application of chemical additives such as acidifiers or urease inhibitors [20,21]. Effective implementation of these strategies requires a sound understanding of how N fractions evolve over the storage period and how internal slurry characteristics relate to gaseous emission rates.

Despite the established importance of pH and temperature in governing NH₃ volatilization, studies that systematically integrate the time-dependent transformation of N fractions with actual emission profiles remain limited. Furthermore, the deployment of direct gas-sensing instrumentation in farm environments is often impractical due to signal drift and high maintenance demands under high-humidity and corrosive conditions. In contrast, pH and temperature are comparatively robust parameters, readily amenable to continuous monitoring, and increasingly incorporated into automated livestock management systems. Against this background, the present study aims to characterize the temporal evolution of nitrogen fractions and the associated NH₃ emission patterns during pig slurry storage in a slurry pit. By establishing a detailed N mass balance and elucidating the relationships between measurable slurry physicochemical properties and gaseous N losses, this work seeks to provide a quantitative basis for improving emission estimation and informing the timing of practical mitigation interventions.

2. Materials and Methods

2.1. Experimental Design for Evaluating Ammonia (NH₃) Emissions

A laboratory experiment was conducted to simulate a storage facility for pig slurry, as depicted in Figure 1. The simulation used a slurry pit consisting of six vessels made of polyvinyl chloride (PVC), each having a diameter of 200 mm and a depth of 500 mm. The vessel was sealed with a plastic lid, and an inlet–outlet was installed on the top of the

cover. Piggery slurry was filled to a height of 300 mm, corresponding to 60% of the total vessel capacity.

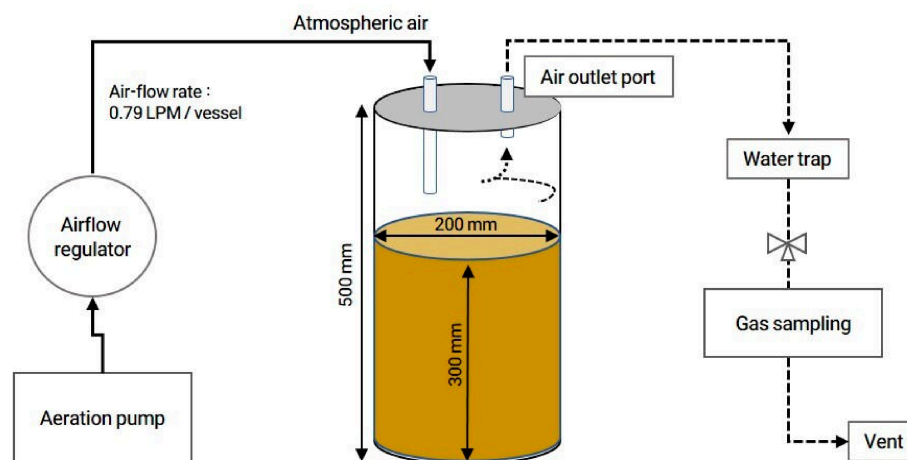


Figure 1. Pig slurry pit simulator designed for measuring NH_3 emissions in the headspace via systematic gas sampling and analysis. Solid arrows indicate the flow path of atmospheric air entering the vessel, while dotted arrows denote the discharge path of air expelled from the vessel to the surrounding environment.

To adhere to the recommended ventilation rate for the pig slurry pit [22,23], the air was exchanged at 7.5 times the headspace volume per hour (ventilation rate: $0.79 \text{ L} \cdot \text{min}^{-1}$). This exchange rate aligns with the lower bound of conditions typically encountered in mechanically ventilated commercial pig houses, where rates have been reported to range from 10.4 to 26.1 h^{-1} [24]. To ensure uniform flow regulation across all six vessels, six air flow meters (LF-101, Unicell Instruments, Bucheon, Republic of Korea) were placed between the inlet tubes connecting the vessels and the air pump (SWT-20, WITHUS AIR, Busan, Republic of Korea).

To prevent moisture formation due to condensation within the exhaust tube connected to the outlet valve of the vessel, a water trap made of a glass bottle was installed upstream of the air discharge point.

The experimental conditions were maintained at a room temperature of 26°C to 35°C throughout the study period. A T-type thermocouple was connected to a module (NI-9213, National Instruments, Austin, TX, USA) to monitor the temperature, and readings were taken every 30 min.

2.2. Pig Slurry Characteristics

Pig slurry was obtained on 1 July 2022 from a sow pig farm ($36^\circ 22' 56'' \text{ N}$ – $126^\circ 56' 56'' \text{ W}$) under the administration of the South Chungcheong Province local authorities. The collected slurry was mixed to prepare representative samples refrigerated at 5°C or lower until analysis. Distilled water, representing the wasted water from waterers and washing water, was evenly mixed with the stored pig slurry before being put into each vessel. The amount of water added was equivalent to approximately 20% of the weight of the pig slurry [25].

The 56-day experiment was conducted from 14 July to 8 September 2022. During this period, pig slurry was sampled from each vessel five times every two weeks. The pH, total solid (TS), volatile solid (VS), total Kjeldahl N (TKN), total ammonium N (TAN), and nitrate N ($\text{NO}_3\text{—N}$) content of the pig slurry underwent analysis utilizing established APHA standard methodologies [26]. Total N (TN) was calculated as the sum of TKN and $\text{NO}_3\text{—N}$. The physico-chemical properties of the raw slurry are presented in Table 1. Free ammonia

nitrogen (FAN) concentration was calculated from measured TAN, pH, and temperature using the equilibrium relationship (Equation (1)):

$$\text{FAN}(\text{g}\cdot\text{kg}^{-1}) = \text{TAN} / (1 + 10^{(pK_a - \text{pH})}) \quad (1)$$

where pK_a is the temperature-dependent dissociation constant of ammonium [27].

Table 1. Physico-chemical properties of raw pig slurry.

Parameter	pH	TS	VS	TN	TKN	TAN	Nitrate-N
Raw pig slurry	7.8 ± 0.1	g·kg ⁻¹ slurry 130.5 ± 2.1	g·kg ⁻¹ slurry 104.5 ± 0.8	g·kg ⁻¹ slurry 8.7 ± 0.1	g·kg ⁻¹ slurry 8.4 ± 0.5	g·kg ⁻¹ slurry 4.7 ± 0.3	g·kg ⁻¹ slurry 0.3 ± 0.1

Values [means ± standard deviation (S.D.), $n = 6$] were measured on a wet basis. TAN represents total ammonium N in pig slurry.

2.3. Measurement of Ammonia (NH₃) Gas Concentration and Emission Rate

The NH₃ concentration was measured using an NH₃ gas sensing module (LGD Compact-A, axetris, Kaegiswil, Switzerland) installed in a real-time monitoring system (OMS-200, Smart Control & Sensing Inc., Daejeon, Republic of Korea). Before the experiment, the gas sensor was calibrated with standard gas. The gas discharged from each vessel was collected in a 10 L Tedlar bag (CEL Scientific Tedlar gas bag, Santa Fe Springs, CA, USA). The gas sample was continuously analyzed using the monitoring system at a flow rate of 1.5 L·min⁻¹ for 5 min followed by a 10 min flushing with ambient air to minimize the risk of sensory overload and signal drift. The NH₃ emission rate was calculated using the following Equation (2) [28]:

$$\text{NH}_3 \text{ emission rate}(\text{g}\cdot\text{d}^{-1}) = (C_E \times V \times \frac{273.15 \times MW}{(273.15 + T_E) \times 22.4 \times 10^3}) - (C_A \times V \times \frac{273.15 \times MW}{(273.15 + T_A) \times 22.4 \times 10^3}) \times \text{UCF} \quad (2)$$

C_E = NH₃ gas concentration of exhausted air (mL·m⁻³);

V = Headspace ventilation rate (m³·min⁻¹);

MW = Molecular weight of NH₃ gas (g·mol⁻¹);

T_E = Exhaust air temperature (°C);

C_A = NH₃ gas concentration of ambient air (mL·m⁻³);

T_A = Ambient air temperature (°C);

UCF = Unit conversion factor (1440 in this study).

2.4. Statistical Analysis

One-way ANOVA and an independent samples t -test (IBM SPSS STATISTICS 22) were used to test for the effect of the storage period on pH, TN, TKN, TAN, FAN in pig slurry, and the daily emissions of gaseous NH₃. A correlation analysis was conducted for gaseous NH₃ and FAN. When the data deviated from a normal distribution, the non-parametric Kruskal–Wallis Test was chosen as the statistical assessment method. Tukey's test was used for post hoc analysis. The analysis of variance was conducted at a significance level of 0.05.

To objectively validate the transition point between the two emission phases, segmented linear regression was applied to all individual replicate observations ($n = 54$) using the 'segmented' package (ver. 2.1-0) in R [29]. A piecewise linear model with a single unknown breakpoint (ψ) was fitted to the daily NH₃-N emission rate as a function of storage time (Equation (3)):

$$E(t) = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot (t - \psi)_+ + \varepsilon \quad (3)$$

where $(t - \psi)_+ = \max(0, t - \psi)$, β_1 represents the slope during Phase I ($t \leq \psi$), and $\beta_1 + \beta_2$ represents the slope during Phase II ($t > \psi$). The breakpoint and its 95% confidence interval were estimated via the iterative algorithm [29]. The Davies test [30] was employed to assess whether a statistically significant change in slope existed within the observation period.

To quantify the reaction kinetics of NH_3 volatilization, a dual-phase regression analysis was applied to the daily emission rates. The storage period was bifurcated into Phase I (Days 1–28) and Phase II (Days 29–56) to evaluate the shift in governing mechanisms. Phase I was fitted to a zero-order kinetic model (constant flux), while Phase II was characterized using a first-order exponential decay model (Equation (4)):

$$E_t = E_0 \times e^{-kt} \quad (4)$$

E_t = NH_3 -N emission rate ($\text{g} \cdot \text{m}^{-2} \cdot \text{d}^{-1}$) at a given storage duration t (days);

E_0 = Initial emission rate at the onset of the decay phase (Day 28 in this study);

k = First-order decay constant (d^{-1});

e = Base of the natural logarithm (Euler's number, 2.718).

3. Results and Discussion

3.1. Nitrogen Transformation in Pig Slurry During Storage

The slurry temperature at the onset of the experiment was 30 °C, rising gradually to a peak of 32 °C on day 28, before declining to 27 °C by day 56. Slurry pH decreased from 7.8 on day 1 to approximately 7.5 by day 56 (Table 2). The gradual decline in pH over the storage period can be attributed, in part, to the progressive loss of alkalinity associated with NH_3 volatilization, in addition to other processes such as VFA accumulation. This trend is likely enhanced under the forced ventilation conditions employed in this study (7.5 headspace volume exchanges per hour), which continuously removed NH_3 from the headspace and maintained the concentration gradient driving surface-layer volatilization.

Table 2. Temporal evolution of pig slurry properties and its diverse nitrogenous compound distribution during the experimental period.

Day	pH	Temperature			Measured Value in This Study				Theoretical Value	
		Mean	Max.	Min.	TN	TKN	TAN	TAN	FAN	FAN
		----- °C -----			---- g·kg ⁻¹ _{slurry} ----		- g·kg ⁻¹ _{slurry} -	- % TN -	- g·kg ⁻¹ _{slurry} -	- % TN -
1	7.8 ± 0.1	29.8	30.6	28.6	8.7 ± 0.6	8.4 ± 0.6	4.7 ± 0.3 ^a	54.0	0.27 ± 0.06 ^a	3.1
14	7.7 ± 0.2	31.0	32.6	28.8	8.1 ± 0.5	7.9 ± 0.4	4.2 ± 0.3 ^{abc}	51.9	0.19 ± 0.05 ^{ab}	2.3
28	7.6 ± 0.1	31.6	32.4	30.1	8.3 ± 0.4	7.7 ± 0.4	4.4 ± 0.2 ^{ab}	53.0	0.20 ± 0.02 ^{ab}	2.4
42	7.5 ± 0.0	29.7	30.1	29.1	7.4 ± 0.3	7.0 ± 0.3	3.5 ± 0.2 ^{bc}	47.3	0.10 ± 0.01 ^b	1.4
56	7.5 ± 0.1	25.7	27.0	24.9	7.5 ± 0.4	6.9 ± 0.4	3.3 ± 0.2 ^c	44.0	0.07 ± 0.01 ^b	0.9

Data (means ± S.E., $n = 6$) with different letters in the same column are significantly different (p -value < 0.05). FAN refers to free NH_3 -N in pig manure, and it represents a theoretical quantity that is determined by the variables of pH and temperature [27].

Over the 56-day storage period, TAN and FAN concentrations declined by 30% and 74%, respectively, relative to their initial values ($p < 0.05$), with statistically significant reductions evident from approximately day 28 onward. The pronounced decline in both fractions during the latter half of the experiment coincides with the observed decrease in slurry temperature after day 28. This thermal dependency is consistent with previous observations that storage at 25 °C for four weeks resulted in no substantial reduction in TN or NH_4^+ levels [31], suggesting that the temperature peak recorded in the present study likely contributed to the accelerated N depletion. This effect can be explained by

a temperature driven shift in the $\text{NH}_4^+ - \text{NH}_3$ equilibrium toward free ammonia (NH_3), coupled with enhanced NH_3 volatilization under forced ventilation, which continuously removed NH_3 from the headspace and promoted sustained nitrogen loss from the slurry.

The mean TAN-to-TN ratio was 53% during the initial 28-day period, decreasing to 46% during the subsequent storage phase (Table 2). These values fall within the reported range of 45–60% for the TAN proportion in raw pig slurry [31,32], and the gradual decline over time is consistent with a progressive nitrogen loss driven primarily by volatilization, relative to ongoing mineralization processes. Theoretical FAN concentrations in pig slurry are determined by the concurrent effects of pH and temperature [13], and FAN represents the primary ammoniacal N fraction responsible for transfer across the slurry–air interface, a partitioning process governed by Henry’s constant [16]. Throughout the experimental period, FAN accounted for 0.9–3.1% of TN, with significantly elevated levels observed during the initial storage phase ($p < 0.05$). FAN concentrations subsequently declined in parallel with TAN, reflecting the progressive depletion of the ammonium pool over time, while still being influenced by the combined effects of pH and temperature.

3.2. Ammonia N ($\text{NH}_3\text{-N}$) Gas Emissions During the Storage Period

$\text{NH}_3\text{-N}$ emission rates ranged from 8.2 to 19.5 $\text{g}\cdot\text{m}^{-2}\cdot\text{d}^{-1}$ and from 30.1 to 65.9 $\text{g}\cdot\text{m}^{-3}\cdot\text{d}^{-1}$ on an area and volume basis, respectively (Figure 2). These values are broadly consistent with previously reported emission rates of 5.3–16.6 $\text{g}\cdot\text{m}^{-2}\cdot\text{d}^{-1}$ for stored pig slurry under comparable conditions [32].

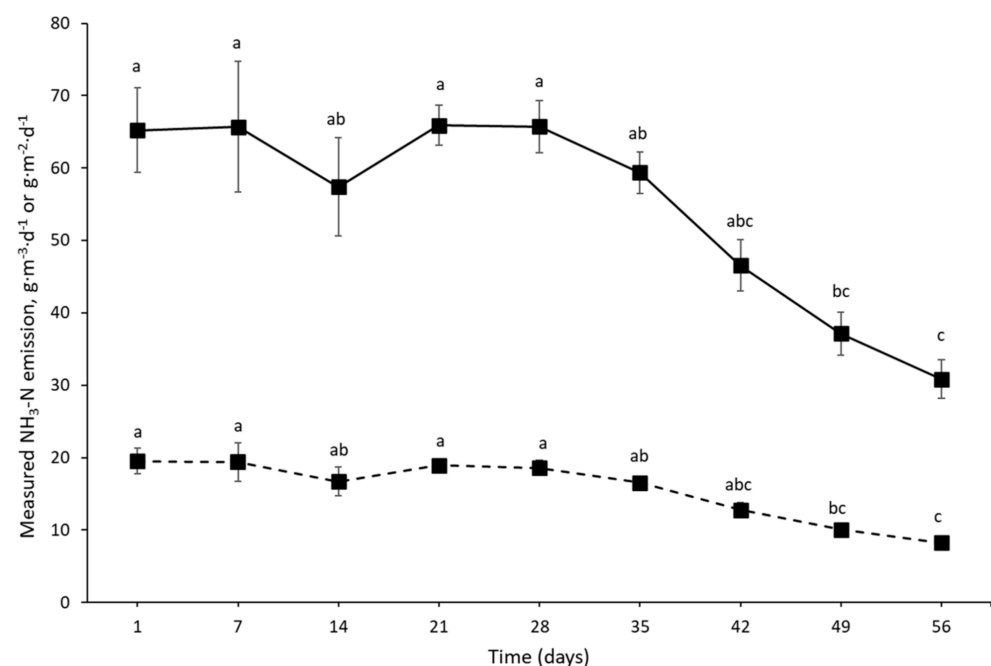


Figure 2. Temporal trends of $\text{NH}_3\text{-N}$ emission rates over a 56-day storage period. The solid and dashed lines represent the $\text{NH}_3\text{-N}$ production per volume of slurry ($\text{g}\cdot\text{m}^{-3}\cdot\text{d}^{-1}$) and unit area ($\text{g}\cdot\text{m}^{-2}\cdot\text{d}^{-1}$), respectively. Different letters on the daily ammonia emission are significantly different (p -value < 0.05). Error bars represent S.E. ($n = 6$).

The temporal emission profile exhibited a distinct two-phase pattern (Figure 3). During Phase I (days 1–28), daily $\text{NH}_3\text{-N}$ emissions remained relatively stable, ranging from 16.7 to 19.5 $\text{g}\cdot\text{m}^{-2}\cdot\text{d}^{-1}$, consistent with zero-order kinetics. While zero-order emission behavior during early storage has been qualitatively associated with high initial TAN availability, the present study provides a quantitative mechanistic basis for this pattern through N fraction mass balance. Specifically, the continuous mineralization of organic N

supplied a net +7.1 g to the TAN pool per vessel (Table 3), effectively compensating for volatilization-driven depletion and sustaining a stable ammonia source throughout Phase I. Mineralization of suspended organic N fractions has been reported to account for up to 19% of initial total organic N during anaerobic pig slurry storage [33]; however, the quantitative linkage between this internal TAN replenishment and the resulting emission kinetics has not been previously established within a slurry storage mass balance framework. This sustained high-flux behavior is attributable to the continuous replenishment of the TAN pool through organic N mineralization, as demonstrated by the mass balance data (Table 3), which effectively prevented the rapid depletion of the ammonia source that would otherwise be expected under progressive volatilization. Consequently, approximately 60% of total cumulative $\text{NH}_3\text{-N}$ emissions (equivalent to $518.6 \text{ g}\cdot\text{m}^{-2}$) were concentrated within the first 28 days of storage (Table 4). This value is broadly comparable to emissions of $256 \text{ g}\cdot\text{m}^{-2}$ reported over 30 days at 20°C [5], with the higher flux in the present study likely attributable to the elevated experimental temperatures.

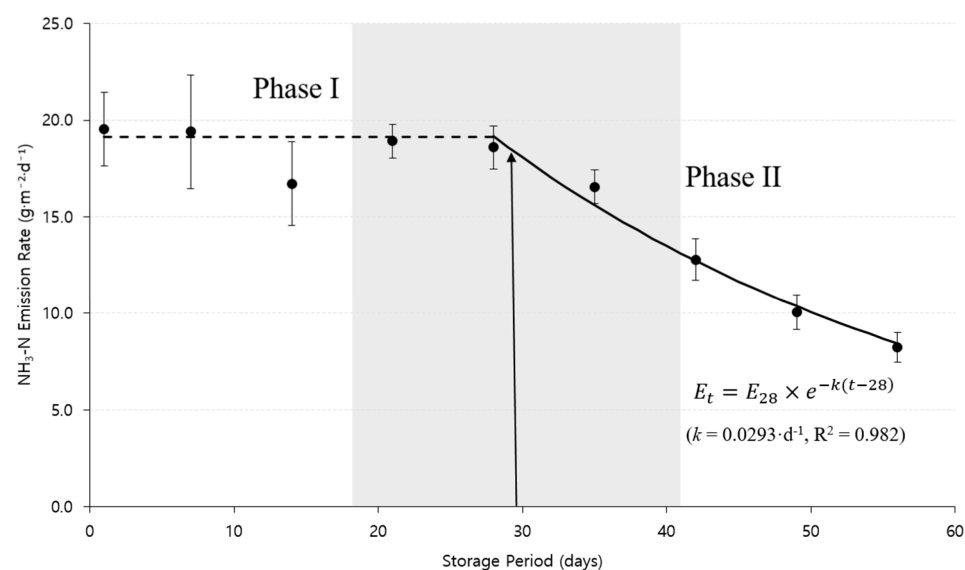


Figure 3. Two-phase kinetic modeling of $\text{NH}_3\text{-N}$ emission rates over a 56-day storage period. Observed data (mean $\pm$ S.E., $n = 6$) are described by a zero-order model during Phase I (Days 1–28, dashed line) and a first-order exponential decay model during Phase II (Days 29–56, solid line; $k = 0.0293 \text{ d}^{-1}$, $R^2 = 0.982$). Segmented regression identified a breakpoint at day 29.9 (black arrow; 95% CI: 18.0–41.8, gray shaded region; Davies test, $p = 0.019$), confirming that the phase transition at day 28 is statistically supported.

To validate the a priori phase division at day 28, segmented regression was applied to all individual observations ($n = 54$). The estimated breakpoint was $\psi = 29.9$ days (95% CI: 18.0 to 41.8 days), encompassing the predetermined boundary of day 28. The Davies test confirmed the existence of a significant slope change across the storage period ($p = 0.019$). The slope during Phase I was not significantly different from zero ($\beta_1 = -0.034 \text{ g}\cdot\text{m}^{-2}\cdot\text{d}^{-1}$ per day, $p = 0.638$), consistent with zero-order kinetics. In contrast, Phase II exhibited a significant negative slope ($\beta_1 + \beta_2 = -0.396 \text{ g}\cdot\text{m}^{-2}\cdot\text{d}^{-1}$ per day, 95% CI: -0.592 to -0.200 , $p < 0.001$), consistent with progressive TAN depletion. The near-zero slope during Phase I reinforces the interpretation that organic N mineralization sustained the TAN pool sufficiently to maintain constant NH_3 flux over the first four weeks. The abrupt transition to a significant negative slope after the breakpoint delineates the point at which mineralization could no longer compensate for volatilization losses, rendering the diminishing TAN pool rate-limiting.

Table 3. Comprehensive N mass balance and transformation over 56 days.

N Pool	Initial (Day 1)	Final (Day 56)	Net Change
Aqueous Phase (slurry)			
TN ($\text{g} \cdot \text{vessel}^{-1}$)	79.6	53.2	28.3
Organic-N ($\text{g} \cdot \text{vessel}^{-1}$)	33.7	25.6	8.1
TAN ($\text{g} \cdot \text{vessel}^{-1}$)	42.8	23.5	19.3
NO ₃ -N ($\text{g} \cdot \text{vessel}^{-1}$)	3.1	4.1	−1.0
Gaseous Phase (emissions)			
Cumulative NH ₃ -N ($\text{g} \cdot \text{vessel}^{-1}$)	0.0	27.1	27.3
Mass Balance			
Total N recovery (%)	-	96.5	-
Unaccounted for N (%)	-	3.5	-

Organic-N is calculated as $[\text{TN} - (\text{TAN} + \text{NO}_3\text{-N})]$. The net change in TN ($28.3 \text{ g} \cdot \text{vessel}^{-1}$) is the sum of daily TN loss $[\sum_{i=1}^n (\text{TN}_{\text{day } i} - \text{TN}_{\text{day } i+1})]$ for 56 days.

Table 4. Cumulative gaseous NH₃-N emissions and reductions in NH₃-N emissions during the storage period.

Day	Cumulative Gaseous NH ₃ -N Emissions		Reductions in NH ₃ -N Emission	
	--- $\text{g} \cdot \text{m}^{-3}$ Slurry ---	--- $\text{g} \cdot \text{m}^{-2}$ Slurry ---	-- % Initial Slurry TN --	-- % Initial Slurry TAN --
1	65.2 ± 6.4 ^a	19.5 ± 1.9 ^a	0.7	1.3
14	888.7 ± 100.6 ^b	262.7 ± 29.7 ^b	9.9	18.4
28	1780.8 ± 105.1 ^c	518.6 ± 31.1 ^c	19.9	36.8
42	2589.1 ± 137.2 ^d	744.3 ± 39.9 ^d	28.9	53.5
56	3119.8 ± 179.0 ^e	888.2 ± 51.3 ^d	34.8	64.4

Data (means ± S.E., $n = 6$) with different letters in the same column are significantly different (p -value < 0.05). Cumulative NH₃-N emissions were estimated using the trapezoidal rule, which assumes linear interpolation between consecutive weekly sampling points.

Following day 28, emissions transitioned into Phase II, characterized by exponential decay ($k = 0.0293 \text{ d}^{-1}$, $R^2 = 0.982$), coinciding with the progressive depletion of TAN and the corresponding decline in theoretical FAN (Table 2). The high goodness of fit of the first-order decay model confirms that the governing emission mechanism shifted from a mineralization-buffered, substrate-replenished regime to one in which the diminishing TAN pool became rate-limiting.

Measured NH₃-N emissions were strongly correlated with theoretical FAN concentrations ($R^2 = 0.74$, Figure 4), supporting the central role of pH and temperature in regulating volatilization potential from untreated stored slurry. This correlation confirms that pH and temperature governed FAN is a primary driver of NH₃ volatilization; however, the remaining 26% of emission variability arises from factors beyond these two parameters. NH₃ emissions are further regulated by a complex set of variables, including wind speed, dry matter content, and surface area [34–36]. In the present study, this residual variability likely reflects mass-transfer resistance at the slurry–air interface and surface-to-bulk TAN concentration gradients that deviate from well-mixed assumptions [13,16]. Consequently, the FAN-based model identifies pH and temperature as dominant predictors of emission potential, while the unexplained variance serves as a quantitative indicator of the additional physical and microbial constraints inherent in stored slurry systems.

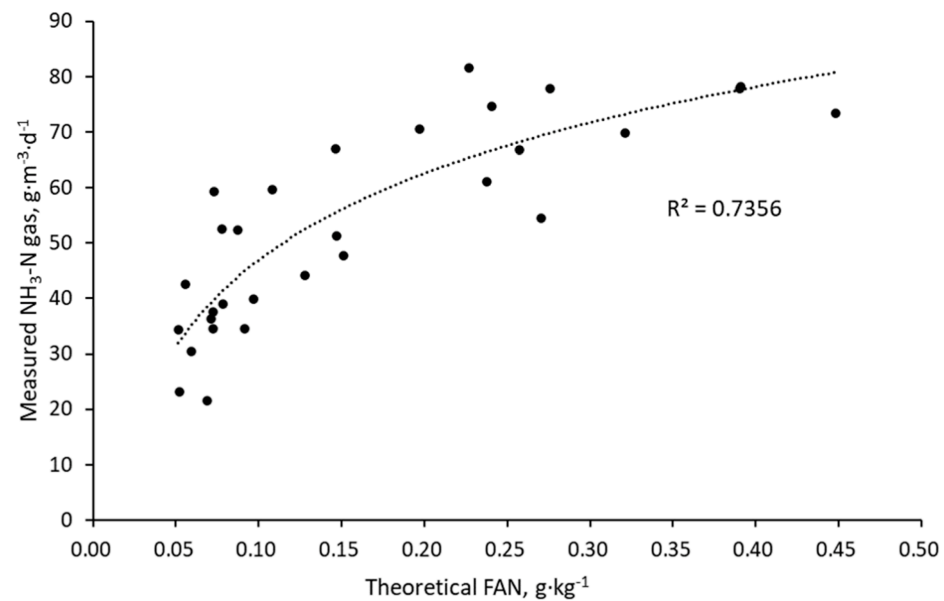


Figure 4. Correlation between theoretical free ammonia nitrogen (FAN) and measured gaseous $\text{NH}_3\text{-N}$ emission rates. Black dots represent individual measured data points ($n = 30$), and the dashed line indicates the best-fit logarithmic regression curve ($y = 22.52 \ln(x) + 98.81$; $R^2 = 0.7356$). A strong positive relationship is confirmed by a Pearson correlation coefficient of 0.806 ($p < 0.0001$).

Over the full 56-day period, cumulative $\text{NH}_3\text{-N}$ emissions accounted for 35% and 64% of the initial TN and TAN, respectively (Table 4). These proportions are somewhat higher than those reported under similar temperature conditions (28% of TN and 53% of TAN) [7], a difference that may partly reflect the higher peak temperatures reached in the present study. Notably, while lowering temperature or pH can suppress NH_3 generation, microbial decomposition of organic matter continues to supply the TAN pool during extended storage, sustaining emission potential well beyond the initial high-flux phase [37]. Under prolonged storage of approximately 43 weeks, TN and TAN have been reported to remain at approximately 32% and 22% of their initial quantities, respectively [31], indicating that long-term N loss through volatilization remains substantial even as daily emission rates decline. A further contributing factor to the observed reduction in NH_3 emissions during Phase II may be the progressive establishment of anaerobic conditions within the stored slurry, which suppresses the activity of aerobic ammonia-releasing microorganisms [16].

The N mass balance (Table 3) provides quantitative evidence of the internal transformation dynamics underpinning the two-phase emission trend. Although total system TN decreased by $28.3 \text{ g} \cdot \text{vessel}^{-1}$, with a corresponding cumulative $\text{NH}_3\text{-N}$ emission of 27.3 g, the TAN pool exhibited a net decrease of only 19.3 g. The remaining 8.1 g reduction in organic N through mineralization, accompanied by a minor 1.0 g increase in $\text{NO}_3^- \text{-N}$ via nitrification, resulted in a net internal contribution of approximately +7.1 g to the TAN pool, effectively buffering the ammonia source against rapid depletion.

The resulting N mass recovery of 96.5% suggests that NH_3 volatilization was likely the predominant N loss pathway under the conditions of this study, although some uncertainty is expected due to the discrete measurement intervals used for emission estimation. The unaccounted-for N fraction (3.5%) falls within the range explainable by gaseous N losses, including N_2O emissions. Reported N_2O fluxes of $0.24 \text{ g} \cdot \text{m}^{-2} \cdot \text{d}^{-1}$ [32], when scaled to the vessel surface area used in this study (0.034 m^2), would account for approximately 1.5% of total TN loss, suggesting that the remainder may be attributable to N_2 gas or other unmeasured pathways. This small residual is further supported by multiple independent studies reporting that N_2O emissions from uncovered, crust-free slurry storage are

negligible. A comprehensive review concluded that N_2O emissions from slurry stores without surface covers are negligible, with the majority of studies reporting fluxes below $0.01 \text{ g} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$ [38]. No significant N_2O emissions were found across three temperature treatments in a pilot-scale slurry storage experiment [7]. At a broader scale, N_2O -N losses from tank-stored pig slurry have been reported to account for only 0.10% of TN, confirming that such emissions are marginal relative to NH_3 volatilization [32]. The low N_2O fluxes observed under these conditions are mechanistically attributable to the predominantly anaerobic environment maintained within undisturbed slurry, which limits the nitrification-denitrification sequence required for substantial N_2O production [39–41]. Nevertheless, as gaseous N species other than NH_3 (i.e., N_2O , N_2 , and NO_x) were not directly measured in the present study, their cumulative contributions cannot be quantitatively excluded, and the potential contributions of denitrification end-products (N_2) and other gaseous intermediates remain unquantified. Future studies incorporating continuous multi-gas monitoring (N_2O , N_2 , and NO_x) would be necessary to fully close the gaseous N balance.

4. Conclusions

This study characterized the temporal relationship between internal N transformations and NH_3 volatilization in stored pig slurry over a 56-day period. A N mass balance with 96.5% recovery, despite uncertainties associated with discrete emission measurements, provided a quantitative basis for interpreting the observed emission patterns and the underlying transformation dynamics.

The central finding is that NH_3 emissions from stored pig slurry deviate from a monotonic decline, instead exhibiting a distinct two-phase trend governed by competing N transformation processes. During the initial 28 days, continuous organic N mineralization supplied a net +7.1 g to the TAN pool per vessel, effectively buffering the ammonia source against the depletion that would otherwise accompany progressive volatilization. This sustained TAN availability maintained stable, high-flux emissions consistent with zero-order kinetics, concentrating approximately 58% of total cumulative NH_3 -N loss within the first month of storage. As the mineralization-driven supply diminished relative to volatilization losses, emissions transitioned into a phase of first-order exponential decay, well described by a decay constant of $k = 0.0293 \text{ d}^{-1}$ ($R^2 = 0.982$).

The strong correlation between measured NH_3 -N emissions and theoretical FAN concentrations ($R^2 = 0.74$) confirms that pH and temperature are the primary determinants of volatilization potential in untreated stored slurry. This relationship further indicates that while theoretical FAN estimates provide a useful upper bound for emission prediction, the actual gaseous flux remains limited by mass-transfer resistance at the slurry–air interface.

From a practical standpoint, the concentration of most N loss within the first month of storage identifies this period as the primary target for mitigation interventions. Strategies such as slurry acidification or pit covering are likely to be most effective when deployed at or before the onset of storage, prior to the accumulation of FAN driven by the temperature-dependent shifts in NH_4^+ – NH_3 equilibrium. The two-phase kinetic framework established here, combined with the demonstrated utility of pH and temperature as predictive variables, offers a foundation for developing low-cost, data-driven emission monitoring tools applicable to automated livestock management systems.

Although this laboratory-scale simulation successfully isolated intrinsic N transformation dynamics and enabled unambiguous identification of the two-phase emission pattern, several experimental constraints must be considered when evaluating the generalizability of these findings. The primary objective of this study was to isolate the intrinsic mechanisms of N transformation and establish a fundamental kinetic model by rigorously controlling variables that introduce complexity under field conditions. As a result, certain factors

known to influence emissions in commercial storage systems, such as oxygen availability, mechanical agitation, and surface crust formation, were deliberately excluded from the experimental design. Specifically, the slurry remained unagitated throughout the 56-day storage period, and no visible crust formation was observed within the sealed vessels. These controlled conditions, together with the fixed ventilation rate of 7.5 h^{-1} representing a conservative lower bound relative to the dynamic airflow encountered in commercial facilities, directly shaped the observed 28-day Phase I plateau and the overall emission magnitudes recorded in this study. The defined temperature range of 26 to 35 °C and the closed batch design, which precluded periodic fresh slurry inputs, further circumscribe the applicable scope of the derived kinetic parameters. Consequently, validation at pilot and farm scales, encompassing a broader range of slurry compositions and climatic conditions, is essential before these findings can be broadly applied. Such field-scale investigations will provide a comprehensive assessment of how physical and chemical factors, including surface crusting and oxygen ingress, interact with and potentially modulate the two-phase emission dynamics established here.

Author Contributions: Conceptualization, H.A. and J.L.; methodology, H.A.; software, J.L.; validation, H.A. and J.L.; formal analysis, J.L.; investigation, J.L.; data curation, J.L.; writing—original draft preparation, J.L.; writing—review and editing, H.A.; visualization, J.L.; supervision, H.A.; project administration, H.A.; funding acquisition, H.A. All authors have read and agreed to the published version of the manuscript.

Funding: This research was supported by Kyungpook National University Research Fund, 2025.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data presented in this study are available on request from the corresponding author.

Acknowledgments: The authors would like to acknowledge JH Shin, YJ Lee, DH Kim, and SH Lee for their assistance in preparing experimental materials and conducting instrument analysis.

Conflicts of Interest: The authors declare no conflicts of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

References

1. Jørgensen, H.; Prapasongsa, T.; Vu, V.T.K.; Poulsen, H.D. Models to quantify excretion of dry matter, nitrogen, phosphorus and carbon in growing pigs fed regional diets. *J. Anim. Sci. Biotechnol.* **2013**, *4*, 42. [\[CrossRef\]](#)
2. Martinez, J.; Guiziou, F.; Peu, P.; Gueutier, V. Influence of Treatment Techniques for Pig Slurry on Methane Emissions during Subsequent Storage. *Biosyst. Eng.* **2003**, *85*, 347–354. [\[CrossRef\]](#)
3. Girard, M.; Palacios, J.H.; Belzile, M.; Godbout, S.; Pelletier, F. Biodegradation in animal manure management. In *Biodegradation-Engineering and Technology*; IntechOpen: London, UK, 2013.
4. Park, S.H.; Lee, B.R.; Jung, K.H.; Kim, T.H. Acidification of pig slurry effects on ammonia and nitrous oxide emissions, nitrate leaching, and perennial ryegrass regrowth as estimated by ^{15}N -urea flux. *Asian-Australas J. Anim. Sci.* **2018**, *31*, 457. [\[CrossRef\]](#) [\[PubMed\]](#)
5. Pereira, J.L.; Perdigão, A.; Marques, F.; Wessel, D.F.; Trindade, H.; Fanguiero, D. Mitigating Ammonia and Greenhouse Gas Emissions from Stored Pig Slurry Using Chemical Additives and Biochars. *Agronomy* **2022**, *12*, 2744. [\[CrossRef\]](#)
6. Rotz, C. Management to reduce nitrogen losses in animal production. *J. Anim. Sci.* **2004**, *82*, E119–E137. [\[PubMed\]](#)
7. Misselbrook, T.; Hunt, J.; Perazzolo, F.; Provolo, G. Greenhouse Gas and Ammonia Emissions from Slurry Storage: Impacts of Temperature and Potential Mitigation through Covering (Pig Slurry) or Acidification (Cattle Slurry). *J. Environ. Qual.* **2016**, *45*, 1520–1530. [\[CrossRef\]](#)
8. Murphy, T.; Cargill, C.; Rutley, D.; Stott, P. Pig-shed air polluted by α -haemolytic cocci and ammonia causes subclinical disease and production losses. *Vet. Rec.* **2012**, *171*, 123. [\[CrossRef\]](#)

9. Preller, L.; Heederik, D.; Boleij, J.; Vogelzang, P.; Tielen, M. Lung function and chronic respiratory symptoms of pig farmers: Focus on exposure to endotoxins and ammonia and use of disinfectants. *Occup. Environ. Med.* **1995**, *52*, 654–660. [\[CrossRef\]](#)
10. Wyer, K.E.; Kelleghan, D.B.; Blanes-Vidal, V.; Schaubberger, G.; Curran, T.P. Ammonia emissions from agriculture and their contribution to fine particulate matter: A review of implications for human health. *J. Environ. Manag.* **2022**, *323*, 116285. [\[CrossRef\]](#) [\[PubMed\]](#)
11. Ravishankara, A.R.; Daniel, J.S.; Portmann, R.W. Nitrous Oxide (N₂O): The Dominant Ozone-Depleting Substance Emitted in the 21st Century. *Science* **2009**, *326*, 123–125. [\[CrossRef\]](#)
12. Philippe, F.X.; Nicks, B. Review on greenhouse gas emissions from pig houses: Production of carbon dioxide, methane and nitrous oxide by animals and manure. *Agric. Ecosyst. Environ.* **2015**, *199*, 10–25. [\[CrossRef\]](#)
13. Chowdhury, A. *Review of Ammonia Emissions from a Pig House Slurry Pit and Outside Storage: Effects of Emitting Surface and Slurry Depth*; Danish Ministry of the Environment, Environmental Protection Agency: Odense, Denmark, 2014.
14. Aneja, V.P.; Chauhan, J.; Walker, J. Characterization of atmospheric ammonia emissions from swine waste storage and treatment lagoons. *J. Geophys. Res. Atmos.* **2000**, *105*, 11535–11545. [\[CrossRef\]](#)
15. Dai, X.-R.; Blanes-Vidal, V. Emissions of ammonia, carbon dioxide, and hydrogen sulfide from swine wastewater during and after acidification treatment: Effect of pH, mixing and aeration. *J. Environ. Manag.* **2013**, *115*, 147–154. [\[CrossRef\]](#)
16. Philippe, F.-X.; Cabaraux, J.-F.; Nicks, B. Ammonia emissions from pig houses: Influencing factors and mitigation techniques. *Agric. Ecosyst. Environ.* **2011**, *141*, 245–260. [\[CrossRef\]](#)
17. Liang, W.-z.; Classen, J.J.; Shah, S.B.; Sharma-Shivappa, R. Ammonia fate and transport mechanisms in broiler litter. *Water Air Soil Pollut.* **2014**, *225*, 1812. [\[CrossRef\]](#)
18. Portejoie, S.; Martinez, J.; Guizieu, F.; Coste, C. Effect of covering pig slurry stores on the ammonia emission processes. *Bioresour. Technol.* **2003**, *87*, 199–207. [\[CrossRef\]](#)
19. Sommer, S.; Christensen, B.; Nielsen, N.; Schjørring, J. Ammonia volatilization during storage of cattle and pig slurry: Effect of surface cover. *J. Agric. Sci.* **1993**, *121*, 63–71. [\[CrossRef\]](#)
20. Regueiro, I.; Coutinho, J.; Fanguero, D. Alternatives to sulfuric acid for slurry acidification: Impact on slurry composition and ammonia emissions during storage. *J. Clean. Prod.* **2016**, *131*, 296–307. [\[CrossRef\]](#)
21. McCrory, D.; Hobbs, P. Additives to reduce ammonia and odor emissions from livestock wastes: A review. *J. Environ. Qual.* **2001**, *30*, 345–355. [\[CrossRef\]](#)
22. MidwestPlanService. *Structures and Environment Handbook*, 11th ed.; Iowa State University: Ames, IA, USA, 1983; Volume 1.
23. Maurer, D.; Koziel, J.; Kalus, K.; Andersen, D.; Opalinski, S. Pilot-Scale Testing of Non-Activated Biochar for Swine Manure Treatment and Mitigation of Ammonia, Hydrogen Sulfide, Odorous Volatile Organic Compounds (VOCs), and Greenhouse Gas Emissions. *Sustainability* **2017**, *9*, 929. [\[CrossRef\]](#)
24. Tabase, R.K.; Van Linden, V.; Bagci, O.; De Paepe, M.; Aarnink, A.J.A.; Demeyer, P. CFD simulation of airflows and ammonia emissions in a pig compartment with underfloor air distribution system: Model validation at different ventilation rates. *Comput. Electron. Agric.* **2020**, *171*, 105297. [\[CrossRef\]](#)
25. Chastain, J.P.; Camberato, J.J.; Albrecht, J.E.; Adams, J. Swine manure production and nutrient content. In *South Carolina Confined Animal Manure Managers Certification Program*; Clemson University: Clemson, SC, USA, 1999; pp. 1–17.
26. American Public Health Association. *Standard Methods for the Examination of Water and Wastewater*; APHA: Washington, DC, USA, 1992.
27. Emerson, K.; Russo, R.C.; Lund, R.E.; Thurston, R.V. Aqueous ammonia equilibrium calculations: Effect of pH and temperature. *J. Fish. Board. Can.* **1975**, *32*, 2379–2383. [\[CrossRef\]](#)
28. Wi, J.; Lee, S.; Kim, E.; Lee, M.; Koziel, J.; Ahn, H. Evaluation of Semi-Continuous Pit Manure Recharge System Performance on Mitigation of Ammonia and Hydrogen Sulfide Emissions from a Swine Finishing Barn. *Atmosphere* **2019**, *10*, 170. [\[CrossRef\]](#)
29. Muggeo, V.M. Estimating regression models with unknown break-points. *Stat. Med.* **2003**, *22*, 3055–3071. [\[CrossRef\]](#)
30. Davies, R.B. Hypothesis testing when a nuisance parameter is present only under the alternative. *Biometrika* **1987**, *74*, 33–43.
31. Popovic, O.; Jensen, L.S. Storage temperature affects distribution of carbon, VFA, ammonia, phosphorus, copper and zinc in raw pig slurry and its separated liquid fraction. *Water. Res.* **2012**, *46*, 3849–3858. [\[CrossRef\]](#)
32. Kupper, T.; Häni, C.; Neftel, A.; Kincaid, C.; Bühler, M.; Amon, B.; VanderZaag, A. Ammonia and greenhouse gas emissions from slurry storage—A review. *Agric. Ecosyst. Environ.* **2020**, *300*, 106963. [\[CrossRef\]](#)
33. Béline, F.; Martinez, J.; Marol, C.; Guiraud, G. Nitrogen transformations during anaerobically stored 15N-labelled pig slurry. *Bioresour. Technol.* **1998**, *64*, 83–88. [\[CrossRef\]](#)
34. Sommer, S.G.; Webb, J.; Hutchings, N.D. New emission factors for calculation of ammonia volatilization from European livestock manure management systems. *Front. Sustain. Food Syst.* **2019**, *3*, 101. [\[CrossRef\]](#)
35. Sommer, S.G.; Hafner, S.D.; Laubach, J.; van der Weerden, T.J.; Leytem, A.B.; Pacholski, A. Model for calculating ammonia emission from stored animal liquid manure. *Biosyst. Eng.* **2022**, *223*, 41–55. [\[CrossRef\]](#)

36. Hafner, S.D.; Pedersen, J.; Fuß, R.; Kamp, J.N.; Dalby, F.R.; Amon, B.; Pacholski, A.; Adamsen, A.P.S.; Sommer, S.G. Improved tools for estimation of ammonia emission from field-applied animal slurry: Refinement of the ALFAM2 model and database. *Atmos. Environ.* **2025**, *340*, 120910. [[CrossRef](#)]
37. Panetta, D.M.; Powers, W.J.; Lorimor, J.C. Management strategy impacts on ammonia volatilization from swine manure. *J. Environ. Qual.* **2005**, *34*, 1119–1130. [[CrossRef](#)]
38. Chadwick, D.; Sommer, S.G.; Thorman, R.; Fanguero, D.; Cardenas, L.; Amon, B.; Misselbrook, T. Manure Management: Implications for Greenhouse Gas Emissions. *Anim. Feed Sci. Technol.* **2011**, *166–167*, 514–531. [[CrossRef](#)]
39. Phillips, V.R.; Holden, M.R.; Sneath, R.W.; Short, J.L.; White, R.P.; Hartung, J.; Seedorf, J.; Schroder, M.; Linkert, K.H.; Pedersen, S.; et al. The Development of Robust Methods for Measuring Concentrations and Emission Rates of Gaseous and Particulate Air Pollutants in Livestock Buildings. *J. Agric. Eng. Res.* **1998**, *68*, 11–24. [[CrossRef](#)]
40. Osada, T.; Rom, H.B.; Dahl, P. Continuous Measurement of Nitrous Oxide and Methane Emission in Pig Units by Infrared Photoacoustic Detection. *Trans. ASAE* **1998**, *41*, 1109–1114. [[CrossRef](#)]
41. Sommer, S.G.; Petersen, S.O.; Søgaard, H.T. Greenhouse Gas Emission from Stored Livestock Slurry. *J. Environ. Qual.* **2000**, *29*, 744–751. [[CrossRef](#)]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.